

UK Horse Racing — Quantitative Strategy Validation

Out-of-sample validation under pre-registered protocol phase6-v1 (fingerprint 76e18ddb6f11f6ad, registered 2026-08-10) · generated 2026-08-10T01:36:48

WARNING — this run did not pass its data-quality gates. RUN ON A DATASET THAT FAILED A PRE-FLIGHT GATE — (data source: the database contains only synthetic races — generated data cannot validate anything about real markets). Results are a harness check, not findings.

1. Executive summary

EDGE_CONFIRMED

The pre-registered decision rule confirms a positive edge. Over the out-of-sample window the primary strategy returned +16.32% across 940 bets, cleared the required significance threshold, and was profitable in the required majority of years. This conclusion was produced by a rule fixed in code before any data was downloaded.

The conclusion above is produced mechanically by a decision rule that was written and content-hashed before any data existed. It cannot be influenced by the results, and no wording in this document may contradict it.

2. Data coverage

Measure	Value
Source	synthetic
Period	2018-01-01 to 2026-06-30
Racing days	3,103
Races	6,206
Runners	49,648
Runners with a price	90.7%
Races with results	100.0%
Out-of-sample rows scored	8,736

Training used 2018-01-01 to 2022-12-31; validation 2023-01-01 to 2024-12-31; and the out-of-sample test window 2025-01-01 to 2026-12-31, with a 14-day embargo between splits. Splits are by date, never at random: a random split would let the model learn from races that had not yet been run.

3. Model performance

Race-level log loss against every baseline. Lower is better. The market row is the only comparison that decides anything — beating a random or uniform forecast is table stakes for any racing model.

forecaster	races	race_log_loss	top1_hit_rate	top3_hit_rate	skill_vs_uniform
model	1092	1.34594	0.4881	0.8407	0.3527
market	1092	1.47951	0.4579	0.793	0.2885

uniform	1092	2.07944	0.12	0.3626	0.0
favourite	1092	2.19257	0.4579	0.6227	-0.0544
random	1092	2.42684	0.0943	0.3498	-0.1671

4. Probability calibration

Ranking runners correctly is not sufficient. Stakes are computed from the probability itself, so a horse quoted at a 20% chance that wins 12% of the time converts a genuine edge into a genuine loss.

Measure	Value
Expected calibration error	0.0091
Bins compared	10

Predicted	Observed	Runners
0.003	0.001	874
0.007	0.003	874
0.016	0.008	873
0.029	0.026	874
0.047	0.038	873
0.076	0.056	874
0.117	0.116	873
0.182	0.183	874
0.277	0.277	873
0.497	0.541	874

5. Betting strategy results

All strategies replayed over identical predictions at 2% slippage and 2% commission. The pre-registered primary hypothesis is **A_ev5 / flat**; every other row is a secondary test and carries a higher significance bar.

Metric	Value
Bets placed	940
Strike rate	39.68%
Average price taken	3.60
Total staked	9,400
Profit	+1,534
Return on investment	+16.32%
Profit factor	1.270

6. Risk analysis

Measure	Value
Starting bankroll	10,000
Final bankroll	11,534
Peak bankroll	11,603

Average stake	10.00
Longest losing streak	10 bets
Sharpe per bet	0.1016
Bet rate (of races seen)	86.08%

Backtested risk is a lower bound on real risk. The simulation does not model account restriction or closure, which is the normal commercial response to a consistently winning customer; it assumes the quoted price is available in the size required; and it does not model a market that moves against a value bettor specifically, which it does.

7. Drawdown analysis

Measure	Value
Maximum drawdown	253
Maximum drawdown (%)	2.23%
Longest losing streak	10 bets

Deepest drawdown periods

start	trough	depth_pct	bets
2025-01-21	2025-02-17	0.02240354605853886	107
2025-11-28	2026-02-10	0.02232995088201418	297
2025-08-12	2025-09-07	0.016080765041691134	61
2025-10-23	2025-11-01	0.008576338145450852	17
2025-03-24	2025-03-28	0.007936236388102157	12

A drawdown observed once in a single out-of-sample window is a sample, not a bound. The deepest drawdown a strategy will experience is almost always still ahead of it.

8. Statistical significance

Quantity	Observed	Required
Bets (primary)	940	200
t-statistic (primary)	+3.11	2.00
Profitable years	100%	50%
Segment bar (Bonferroni, 12 tests)	2.89	-

Does the model know anything the price does not?

model adds information beyond the price ($b_{\text{model}} = +0.785$, $z = 6.02$)

Term	Coefficient	z
Market price	+0.5088	+2.74
Model	+0.7850	+6.02

This is a forecast encompassing regression. Because the bookmakers' own implied probability is already in the equation, the model's coefficient measures only what it adds *given that the price is already known*. It is a considerably harder test to pass than beating the market on log loss, and it is the one worth reporting.

9. Final verdict

EDGE_CONFIRMED

Positive edge confirmed: +16.32% ROI over 940 bets ($t = 3.11$)

Reasoning, in the order the frozen rule evaluates it:

1. primary ROI +16.32% over 940 bets
2. primary $t = +3.11$ (threshold 2.0)
3. profitable years 100% of 2 (threshold 50%)
4. model beats the market baseline on race log loss (-0.1336)

This document reports a research result. It is not financial advice and not a recommendation to stake money. Betting markets are efficient enough that most apparent edges are overfitting, stale prices, or unmodelled costs; never stake money you cannot afford to lose.